

Flip: Driving innovation for hypergrowth with cloud migration

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Fintech scaleup Flip migrated its cloud collaboration and infrastructure to AWS Cloud, leaving more room for innovation while managing infrastructure, delivering vastly improved effective performance, and its teams and customers.

ABOUT FLIP



About Flip

Founded in 2015, Flip is a fintech-based payments platform serving more than 10 million users and integrated with hundreds of businesses in Indonesia. A leading innovator in the region, Flip has driven free interbank transfers and low-cost transactions for its users, helping them achieve their financial goals by delivering the fairest financial products and services available.

Industries: Financial Services

Google Cloud results

- Reduces data and analytics update cycles from days to minutes with BigQuery
- Speeds up data and analytics insights for daily job runs from 8 hours to 35 seconds
- Peak 'pay day' events are now a more seamless event for engineers
- Strengthens infrastructure security by improving security posture underpinned by Google-

Workload
migration
increased
security
scalability
reliability
performance

Location: Indonesia

Cloud SQL (<https://cloud.google.com/sql/>),

Google Cloud (<https://cloud.google.com/>)

Apigee API Platform

(<https://cloud.google.com/apigee/>)

Cloud IAM (<https://cloud.google.com/iam/>).

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Load Balancing

(<https://cloud.google.com/load-balancing/>)

Cloud SQL (<https://cloud.google.com/sql/>) 'gmail

Compute Engine

(<https://cloud.google.com/compute/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

, Docs

Cloud IAM (<https://cloud.google.com/iam/>) locs/

Cloud Memorystore

(<https://cloud.google.com/memorystore/>)

Google Workspace

(<https://workspace.google.com/>)

Gmail

(<https://workspace.google.com/products/gmail/>)

Calendar

(<https://workspace.google.com/products/calendar/>)

Cloud Identity-Aware Proxy

Google Chat

(<https://workspace.google.com/products/chat/>)

Docs

(<https://workspace.google.com/products/docs/>)

relax your challenge. We're here to help.

grade
security
• Creates a
positive
impact on
customer
experience

Google Drive

(<https://workspace.google.com/products/drive/>)

Google Meet

(<https://workspace.google.com/products/meet/>)

Sheets

(<https://workspace.google.com/products/sheets/>)

Slides

(<https://workspace.google.com/products/slides/>)

Cloud Identity-Aware Proxy

(<https://cloud.google.com/iap/>)

When money becomes data, performing digital transactions should be simple no matter where you live and who you want to pay. That's exactly the

driving mission at Flip, an Indonesian fintech innovator that has been working to create fair financial solutions by delivering free and low-cost transactions to consumers and businesses since it was licensed by Bank Indonesia.

"There are hundreds or thousands of financial products in Indonesia," says Henry Suryawirawan, VP of Engineering at Flip. "However, transferring money between different banks has been a big hassle and incurs charges. We now enable everyone, B2C and B2B, to make single or bulk transfers easily."

Before Flip, transactions would cost 6,500 Rupiah (\$0.42 USD) for every domestic transfer. But through Flip, customers can transfer funds for free. As of 2022, Flip serves over 10 million users as well as hundreds of companies, allowing everything from domestic and international money transfers, e-wallet services to phone credit top-ups, paying electricity bills, financial management for businesses, and more.

As Flip has moved from startup to scaleup, the business found itself with growing pains on its original infrastructure platform. With a growing user base, number of products and innovations for its users, it noticed less agility on its infrastructure, so Flip began to explore new options for its cloud delivery platform.

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Consolidating collaboration through Google Workspace

Flip's first step into working with Google came through a migration to Google Workspace. With a wide range of collaboration and document storage tools used across its teams, Flip decided it needed to consolidate its office tools into a single solution and found Google Workspace to be the best fit.

"We polled the team internally and found the majority preferred to use Google Workspace," says Henry. "We've found that collaboration has become seamless, and features like Gmail

(<https://workspace.google.com/products/gmail/>) and

Google Calendar

(<https://workspace.google.com/products/calendar/>) just work so well. We've also been able to better govern knowledge management through Google Drive

(<https://workspace.google.com/products/drive/>), Google Docs (<https://workspace.google.com/products/docs/>) and related services."

With this baseline for the organization, Flip also found that preparing to migrate its wider cloud infrastructure into Google Cloud became an easy choice with Identity and Access Management

(<https://cloud.google.com/iam>) features already in place across the business.



Effective real-time insights made possible with BigQuery

As the first step on its Google Cloud journey, Flip moved its data warehouse to BigQuery.

(<https://cloud.google.com/bigquery/>) to improve its data analysis processes.

"Previously, data queries took a very long time," says Henry. "With such large databases, it was also becoming a source of resource contention, with applications, engineers and data teams all trying to query the data at once."

"This first project unlocked a lot of capabilities for our data and analytics team. They built a single source of truth, a dashboard to reflect data on our users, revenue growth, and can even show fraud analysis. What took hours or even days before was now updated in minutes."

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With improved insights, the team found many decisions could be made faster than ever before, and innovation was unleashed.

"When we use BigQuery, we can make faster decisions in just minutes on which products to improve, which customer experiences need improvement, and even what new products we could build," says Henry. "And being serverless meant we

didn't need to involve our infrastructure team. The costs are now per query, which is also a benefit to the business."

Flip also adopted Datastream

(<https://cloud.google.com/datastream>) early on in this process to keep BigQuery data fresh and gain insights closer to real-time than ever before.

"The Google Cloud team has been very supportive," says Henry. "From the Workspace migration through to offering access to Datastream while it was in Preview. If we ever had any hiccups, they always helped us."

Easier load scaling on peak days with Google Cloud

Flip continued its migration of wider applications workloads to Google Cloud (<https://cloud.google.com/>).

The company is now running on tools such as

Kubernetes

(<https://cloud.google.com/kubernetes-engine/>), Compute

Engine (<https://cloud.google.com/compute/>), Cloud SQL

(<https://cloud.google.com/sql/>), Memorystore

(<https://cloud.google.com/memorystore/>), and Cloud

Load Balancing

(<https://cloud.google.com/load-balancing/>).

"Google Cloud is one of the friendlier clouds to work with," says Henry. "API and SDK support is simple, there's good documentation, and whenever there's a new product or service, the APIs are ready. In terms of experience, everything is just great for developers."

"We used to have reliability issues that required us to guard the infrastructure carefully on such peak days. With Google Cloud, we can focus less on maintenance, as it just scales so much better now. In a sense, it's a more peaceful event for the team now."

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Overall, Henry says he has also found the performance to be much easier to manage than its previous platform. This difference has been most

noticeable in the management of peak 'pay days' during the month.

"There are two peak days in the month when people receive their salary and send money at the same time," says Henry. "We used to have reliability issues that required us to guard the infrastructure carefully on such peak days. With Google Cloud, we can focus less on maintenance, as it just scales so much better now. In a sense, it's a more peaceful event for the team now."

He goes on to add, "We like that Google Cloud is battle tested through Google's own products. It's that proven performance and reliability that enticed us to move to this platform."

Strengthened infrastructure security with Identity-Aware Proxy

Flip also took the migration cycle as an opportunity to strengthen the company's security posture.

Moving away from its previous platform was a chance for the fintech company to move beyond its startup roots and instill tighter security rules, better governance, and ensure the right people have the right access.

Underpinned by Google-grade security, Flip is also now using [Identity-Aware Proxy](#)

(<https://cloud.google.com/iap/>) to manage secure access to its infrastructure. Henry says it is now used as a superior access management option over VPN, as well as for securing legacy applications that can't integrate with SSO.

Despite all the changes to the infrastructure, there has been no interruption to the customer experience for Flip's users. In fact, Henry believes that reliability has only been improved. Any signs of slow performance or downtime are now in the past.

Ultimately, with the right tools now in place to gain faster insights and make decisions more efficiently, Flip is able to generate more improvements to its customer experience in a shorter cycle than ever before.

Modernizing to take full advantage of serverless technologies

With almost all workloads now on Google Cloud, Henry says that Flip feels more secure in the pursuit of its mission "to provide the fairest financial services to millions of Indonesian people and businesses," as it continues to advance its infrastructure through Google Cloud.

"Scaling is one thing, but then it's also modernization," says Henry. "Now that we've successfully migrated, we're looking to adopt more native Google services and serverless technologies."

Cloud Spanner (<https://cloud.google.com/spanner>), Pub/Sub (<https://cloud.google.com/pubsub>) and Traffic Director (<https://cloud.google.com/traffic-director>), are all on Flip's radar for ways to deepen its use of advanced Google Cloud services.

"The natural progression will also be in machine learning," says Henry. "Our data team is looking into

that area right now. We aspire for more scale, more users, and more products. We feel confident that Google infrastructure will support that growth."